A Literature Survey of Hybrid Big Data and Quantum Computing for Smart Delivery Route Optimization

J. Jayanth, B. Kaivalya, P. Shyam Sai
Department of Computer Science and Engineering
SRM University-AP, Amaravati, Andhra Pradesh, India

Prof. Vishnu Prasad Varma
Department of Computer Science and Engineering
SRM University-AP, Amaravati, Andhra Pradesh, India

***Abstract*—The increasing volume of transportation and delivery data has created a requirement for efficient route optimization methods. Vehicle routing problems become difficult when traffic conditions, customer demands, vehicle constraints, delivery time windows, and other transportation conditions change dynamically. Big Data analytics provides methods for processing large transportation datasets, while quantum computing provides emerging methods for combinatorial optimization.**

This literature survey reviews 20 selected studies related to Big Data, logistics, intelligent transportation systems, vehicle routing, quantum computing, route optimization, and dynamic transportation. The selected studies are organized according to their primary problem categories: Big Data, Logistics, Route Optimization, Quantum Computing, and Dynamic Transportation. The survey examines the use of Big Data for transportation information processing and the application of quantum optimization methods, particularly the Quantum Approximate Optimization Algorithm and Quadratic Unconstrained Binary Optimization formulation, to routing problems. The reviewed literature is used to identify research directions for a hybrid Big Data and quantum computing approach to smart delivery route optimization.

***Index Terms*—Big Data, Quantum Computing, Vehicle Routing Problem, QAOA, Smart Logistics, Intelligent Transportation Systems, Route Optimization, Last-Mile Delivery**

I. Introduction

The growth of logistics and delivery services has increased the importance of efficient vehicle routing and transportation optimization. Delivery route planning involves determining suitable routes for vehicles while considering customer requirements, vehicle capacities, travel distances, traffic conditions, and delivery constraints.

Transportation systems generate large amounts of information from GPS devices, traffic monitoring systems, sensors, vehicles, mobile devices, and other connected transportation sources. Big Data analytics can be used to collect, process, and analyze such information for intelligent transportation applications [3].

Chen [1] investigated cold-chain logistics distribution optimization using Big Data analysis and real-time traffic information. The study considered vehicle routing with customer demands and soft time-window constraints and applied a parallel coarse-grained genetic algorithm.

Zheng [2] studied the Vehicle Routing Problem using a Big Data analytic approach. The study considered dynamic traffic flow, changing customer demand, multiple depots, and soft time windows. Traffic information was updated periodically to support dynamic route optimization.

Quantum computing has also been investigated for transportation optimization. Azad *et al.* [7] studied the Vehicle Routing Problem using the Quantum Approximate Optimization Algorithm. The routing problem was formulated for quantum optimization and evaluated using Qiskit and quantum computing experiments.

Yarkoni *et al.* [8] investigated quantum computing for traffic navigation using live traffic information and a D-Wave quantum processor.

The objective of this literature survey is to examine research related to Big Data and quantum computing for transportation and logistics optimization and to identify how the two areas can be combined for smart delivery route optimization.

II. Survey Methodology

This literature survey considers 20 selected research papers covering Big Data, logistics, intelligent transportation systems, vehicle routing, quantum computing, and dynamic transportation.

The selected papers were reviewed according to their research problem, application area, optimization method, and relevance to smart delivery route optimization.

The papers were organized into five primary categories:

1) Big Data
2) Logistics
3) Route Optimization
4) Quantum Computing
5) Dynamic Transportation

The classification is used to organize the selected literature. Several papers discuss more than one research area, but each paper is assigned to one primary category for the purpose of the distribution analysis.

The publication years of the selected papers range from 2018 to 2026.

III. Publication Timeline

The selected literature covers the period from 2018 to 2026. The number of selected papers varies across the publication years.

The selected papers are distributed as follows:

- 2018: 1 paper
- 2019: 3 papers
- 2020: 2 papers
- 2021: 3 papers
- 2022: 3 papers
- 2023: 2 papers
- 2024: 0 papers
- 2025: 5 papers
- 2026: 1 paper

Figure 1 presents the number of selected papers collected for each publication year.

IV. Proposed Smart Delivery Route Optimization Workflow

The literature reviewed in this survey provides the main components for a smart delivery route optimization framework. Transportation information can be collected from GPS devices, traffic information, customer requirements, vehicle information, weather conditions, and road conditions.

The collected information can be processed using Big Data techniques to obtain transportation information required for route optimization. The resulting transportation information can be used to construct a Vehicle Routing Problem.

The routing problem can subsequently be represented using binary decision variables and transformed into a QUBO formulation. Quantum optimization methods such as QAOA can then be investigated for solving the resulting optimization problem.

Classical processing remains necessary for data preparation, constraint handling, solution interpretation, feasibility checking, and reconstruction of the final delivery routes.

V. Big Data in Logistics

Big Data analytics is used in transportation and logistics to process large volumes of information generated by transportation systems. The reviewed literature considers traffic information, vehicle information, customer requirements, logistics information, and other transportation data.

Chen [1] investigated cold-chain logistics distribution optimization using Big Data analysis. The study incorporated real-time traffic information and considered delivery requirements, vehicle routing, customer demand, and soft time windows.

Zheng [2] investigated a Big Data analytic approach for the Vehicle Routing Problem. The approach considered dynamic traffic flow, customer demand, multiple depots, and soft time windows.

Cuzzocrea *et al.* [5] investigated a Big Data analytics framework for logistics problems in smart-city environments. The work considered logistics planning and delivery problems in an urban environment.

Zhu *et al.* [3] presented a survey of Big Data analytics in Intelligent Transportation Systems. The study discussed transportation data sources, data collection, data analytics, and applications such as traffic flow prediction and route planning.

These studies demonstrate the relevance of Big Data processing to transportation and logistics applications.

VI. Transportation Data and Route Optimization

Transportation route optimization depends on information about vehicles, customers, roads, traffic, and travel conditions.

Zhu *et al.* [3] reviewed transportation data collection and analytics in Intelligent Transportation Systems. The study considered data generated from multiple transportation sources and discussed applications of Big Data analytics.

Issaoui *et al.* [4] proposed an advanced system for delivery operations in a smart logistics environment. The study considered delivery prediction, dynamic coordination, and path optimization.

Hussain [6] discussed artificial intelligence-based route optimization for a sustainable logistics ecosystem. The study considered real-time information from IoT and GPS systems together with traffic and weather information.

The reviewed studies indicate that route optimization can benefit from transportation information that is continuously collected and processed.

VII. Vehicle Routing Problems

The Vehicle Routing Problem involves determining routes for a set of vehicles serving customers while satisfying relevant constraints. Different applications introduce additional conditions such as vehicle capacity, customer demand, time windows, multiple depots, and changing transportation conditions.

Zheng [2] addressed a dynamic Vehicle Routing Problem using Big Data analytics. The study considered changing traffic flow and customer requirements.

Azad *et al.* [7] investigated the Vehicle Routing Problem using QAOA. The routing problem was formulated as a quantum optimization problem and evaluated using Qiskit.

Yarkoni *et al.* [8] investigated quantum computing for traffic navigation using live traffic information.

Acuña Acuña *et al.* [12] investigated quantum-enhanced cognitive modeling for logistics route optimization and applied QAOA to the optimization process.

VIII. Quantum Computing for Transportation

Quantum computing has been investigated as an approach for optimization problems in transportation.

Wang *et al.* [10] reviewed the application of quantum computing in Intelligent Transportation Systems. The study discussed applications including path planning, transportation operation management, and facility layout.

Sarkar *et al.* [9] reviewed quantum computing applications in Intelligent Transportation Systems using publications from the period 2014–2025. The review discussed transportation optimization, vehicle routing, demand forecasting, vehicular security, and autonomous intelligence.

The reviewed quantum transportation studies demonstrate the use of quantum optimization approaches for transportation-related problems while also identifying limitations associated with current quantum systems.

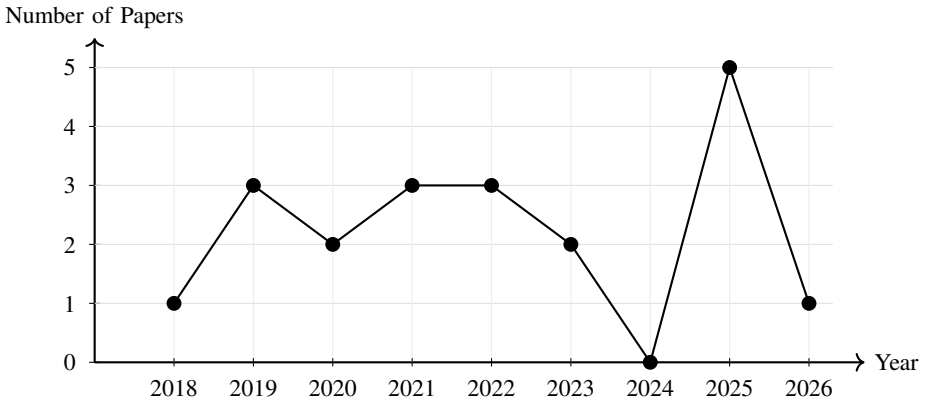


Fig. 1. Number of selected papers collected for each publication year from 2018 to 2026.

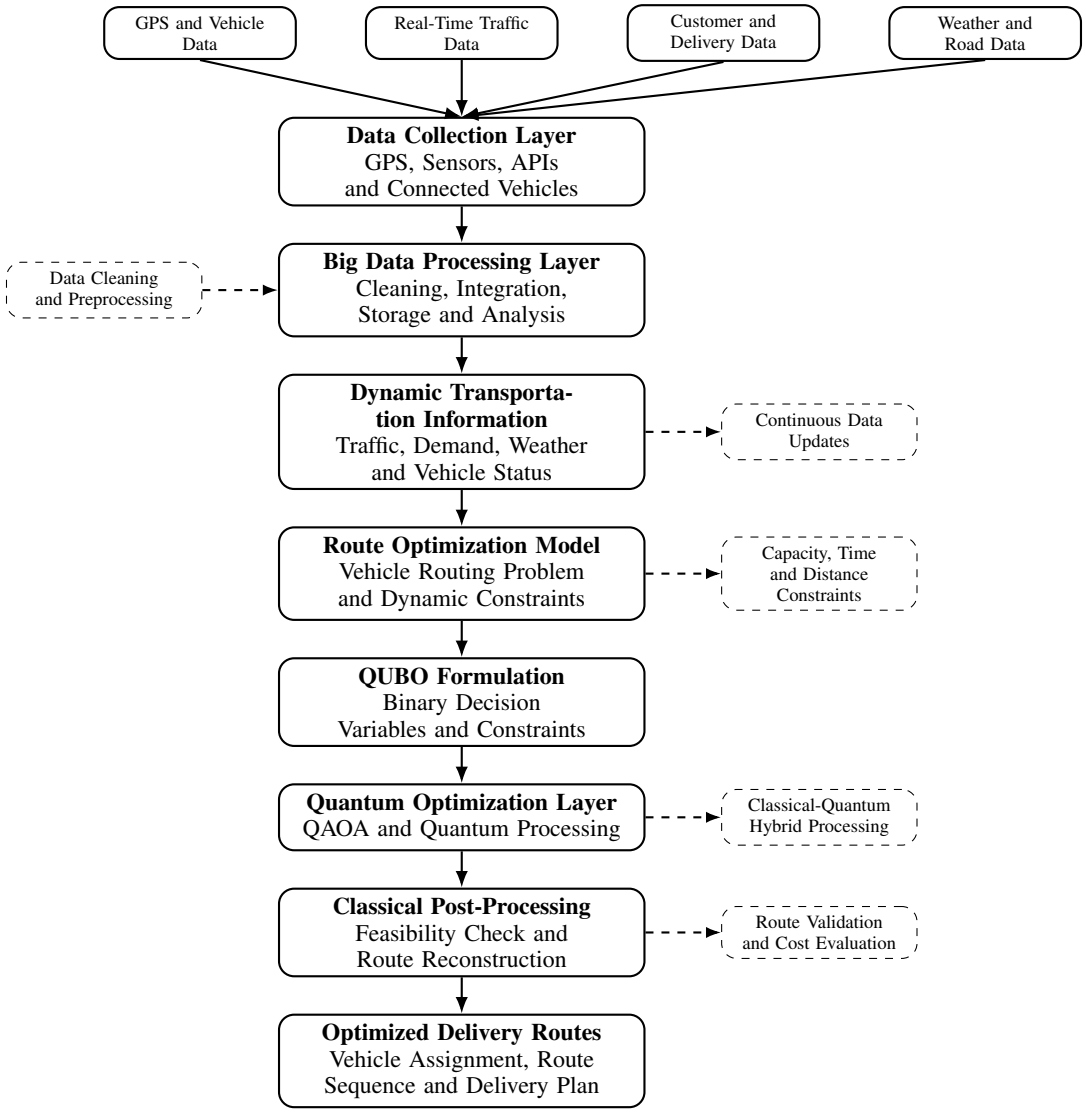


Fig. 2. Proposed hybrid Big Data and quantum computing workflow for smart delivery route optimization.

IX. QAOA and Quantum-Based Smart Logistics

The Quantum Approximate Optimization Algorithm is a hybrid quantum-classical algorithm that can be applied to combinatorial optimization problems.

For route optimization, a routing problem can be represented

using binary decision variables. The resulting optimization problem can be formulated as a Quadratic Unconstrained Binary Optimization problem and subsequently processed using a quantum optimization algorithm.

The relationship between the major stages considered in the survey can be represented as:

$$\text{Big Data} \rightarrow \text{Traffic Information} \rightarrow \text{Vehicle Routing Problem} \rightarrow \text{QUBO} \rightarrow \text{QAOA} \tag{1}$$

Azad *et al.* [7] investigated QAOA for the Vehicle Routing Problem and compared the quantum approach with classical optimization.

El Azzaoui *et al.* [11] proposed a QAOA-based approach for smart logistics and considered scalability, security, cost, time, and carbon-related objectives.

Acuña Acuña *et al.* [12] investigated quantum-enhanced cognitive modeling for logistics route optimization using QAOA.

X. Big Data Applications Related to Transportation

Several selected studies apply Big Data techniques outside direct delivery routing but provide methods relevant to transportation data processing and optimization.

Seyyedsalehi and Khansari [13] investigated Virtual Machine Placement Optimization for Big Data applications. The study considered resource utilization and optimization in Big Data processing environments.

Elhoseny *et al.* [14] proposed a hybrid Internet of Things model for managing Big Data in health service applications. The study investigated optimization methods for resource management.

Jinadu *et al.* [15] investigated distributed database system optimization for Big Data applications. The study considered distributed storage, query optimization, replication, transaction management, and service delivery.

These studies provide supporting perspectives on the processing and management of large datasets that can be relevant when handling large transportation datasets.

XI. Dynamic Transportation Applications

Transportation problems can change while an operation is in progress. Traffic conditions, customer requirements, weather conditions, and transportation resources may change over time.

Niyazi and Behnamian [16] investigated a three-stage dynamic model for disaster relief logistics and wounded transportation. The study used transportation information and online updates to support changes in routing decisions.

Thai and Huh [17] investigated patient transportation using Big Data analysis and transportation information. The study used public transportation data and route-related information for patient transportation.

These studies demonstrate the importance of dynamic information in transportation decision-making.

XII. Last-Mile Delivery Optimization

Last-mile delivery involves the final transportation stage between a logistics facility and the customer. Route planning in this stage is affected by traffic, road conditions, customer requirements, vehicle availability, and other environmental factors.

Jagarlamudi [18] investigated computer vision and IoT for optimized last-mile delivery in smart logistics systems. The study combined information from connected devices with computer vision and optimization methods.

The work considered a delivery scenario involving multiple vehicles and packages and investigated the effect of the proposed approach on delivery time, fuel consumption, and carbon emissions.

Paper [19] in the selected collection has the same title as paper [18] and is therefore retained with the same title in the survey collection.

XIII. Dynamic Truck-UAV Delivery Routing

Collaborative truck-UAV delivery is another logistics routing problem in which multiple transportation resources are coordinated.

Yuan *et al.* [20] investigated collaborative truck-UAV delivery routing under dynamic weather conditions and customer demands. The study considered wind conditions and changing customer demand in the delivery routing problem.

The study used adaptive optimization methods for collaborative truck-UAV routing and considered the effect of dynamic transportation conditions.

XIV. Quantitative Analysis of the Selected Literature

The selected 20 papers were classified into five primary problem categories for analysis.

The category distribution is:

- Big Data: 4 papers
- Logistics: 8 papers
- Route Optimization: 4 papers
- Quantum Computing: 3 papers
- Dynamic Transportation: 1 paper

The distribution is based on the primary problem assigned to each paper for this literature survey.

XV. Category-wise Publication Distribution

The selected papers were also organized according to their primary problem category and publication year. Table **??** shows the number of papers belonging to each category for each publication year.

XVI. Details of the Selected Literature

Table I lists the 20 selected papers used in this literature survey.

XVII. Papers Organized by Problem

The selected papers are organized according to their primary problem category.

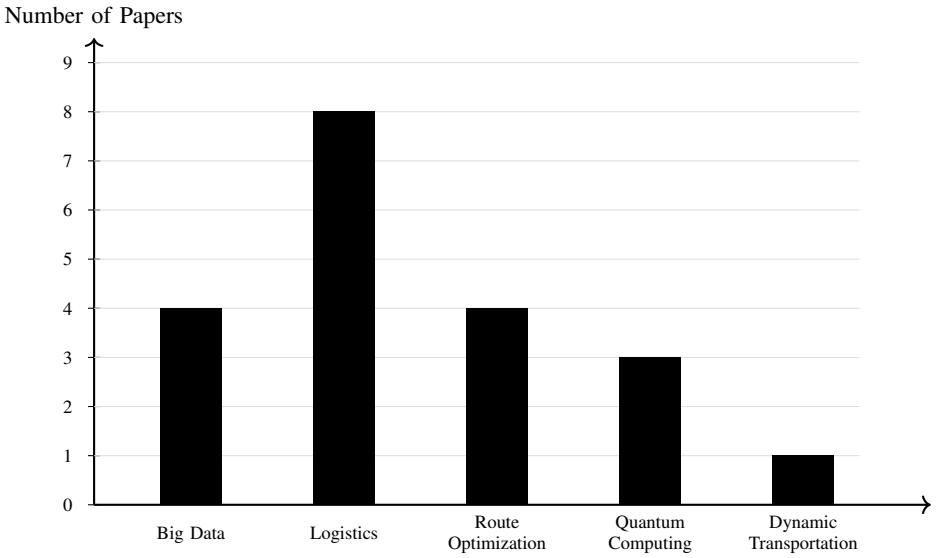
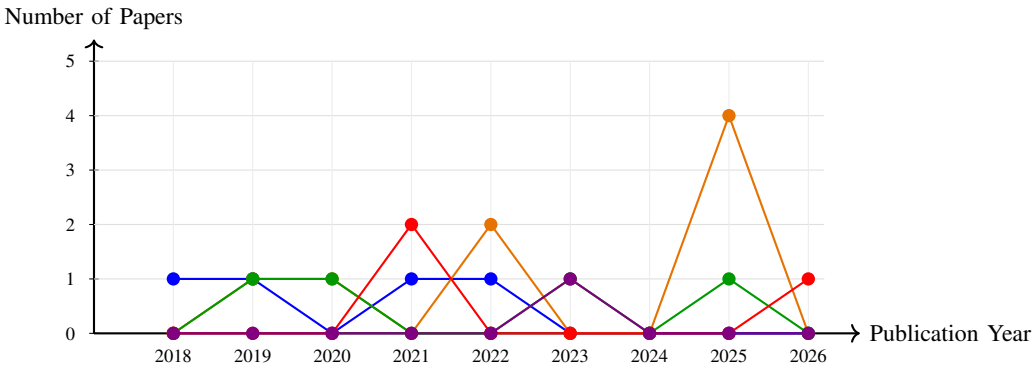


Fig. 3. Distribution of the 20 selected papers according to their primary problem category.



Category	2018	2019	2020	2021	2022	2023	2024	2025	2026
Big Data	1	1	0	1	1	0	0	0	0
Logistics	0	1	1	0	2	0	0	4	0
Route Optimization	0	1	1	0	0	1	0	1	0
Quantum Computing	0	0	0	2	0	0	0	0	1
Dynamic Transportation	0	0	0	0	0	1	0	0	0

Fig. 4. Timeline of the 20 selected papers for the literature survey, divided according to their primary problem category.

XVIII. Comparative Analysis

The selected studies cover different aspects of transportation and logistics optimization.

A. Big Data

The Big Data studies focus on the collection, processing, and analysis of large transportation datasets. Chen [1], Zheng [2], and Zhu *et al.* [3] investigate the use of transportation data for logistics and Intelligent Transportation Systems.

B. Logistics

The logistics studies address delivery operations, logistics planning, transportation services, and last-mile delivery [4]–[6], [17], [18], [20].

TABLE I
DETAILS OF THE 20 SELECTED PAPERS.

ID	Reference or Study	Year	ID	Reference or Study	Year
[1]	Intelligent Algorithms for Cold Chain Logistics Distribution Optimization Based on Big Data Cloud Computing Analysis	2020	[11]	A Quantum Approximate Optimization Algorithm Based on Blockchain Heuristic Approach for Scalable and Secure Smart Logistics Systems	2021
[2]	Solving Vehicle Routing Problem: A Big Data Analytic Approach	2019	[12]	Quantum-Enhanced Cognitive Modeling for Advanced Logistics Route Optimization	2025
[3]	Big Data Analytics in Intelligent Transportation Systems: A Survey	2019	[13]	Virtual Machine Placement Optimization for Big Data Applications in Cloud Computing	2022
[4]	An Advanced System to Enhance and Optimize Delivery Operations in a Smart Logistics Environment	2022	[14]	A Hybrid Model of Internet of Things and Cloud Computing to Manage Big Data in Health Services Applications	2018
[5]	A Big-Data-Analytics Framework for Supporting Logistics Problems in Smart-City Environments	2019	[15]	Distributed Database System Optimization for Improved Service Delivery in Mobile and Cloud BigData Applications	2021
[6]	Revolutionizing Route Optimization Systems with Artificial Intelligence for a Smarter, Sustainable Logistics Ecosystem	2025	[16]	Application of Cloud Computing and Big Data in Three-Stage Dynamic Modeling of Disaster Relief Logistics and Wounded Transportation	2023
[7]	Solving Vehicle Routing Problem Using Quantum Approximate Optimization Algorithm	2023	[17]	Optimizing Patient Transportation by Applying Cloud Computing and Big Data Analysis	2022
[8]	Quantum Shuttle: Traffic Navigation with Quantum Computing	2020	[18]	Leveraging Computer Vision and IoT for Optimized Last-Mile Delivery in Smart Logistics Systems	2025
[9]	Integrating Quantum Computing into Intelligent Transportation Systems	2026	[19]	Leveraging Computer Vision and IoT for Optimized Last-Mile Delivery in Smart Logistics Systems	2025
[10]	Shaping the Future of the Application of Quantum Computing in Intelligent Transportation System	2021	[20]	Collaborative Truck-UAV Delivery Routing Optimization Under Dynamic Weather Conditions and Customer Demands	2025

TABLE II
SELECTED PAPERS ORGANIZED ACCORDING TO THEIR PRIMARY PROBLEM.

Problem	Paper IDs	Total
Big Data	[3], [13], [14], [15]	4
Logistics	[1], [4], [5], [6], [17], [18], [19], [20]	8
Route Optimization	[2], [7], [8], [12]	4
Quantum Computing	[9], [10], [11]	3
Dynamic Transportation	[16]	1
Total	**20 papers**	**20**

C. Route Optimization

Vehicle routing and route optimization studies consider the determination of efficient routes under transportation constraints. These include dynamic traffic, customer demand, delivery requirements, and routing constraints [1], [2], [7].

D. Quantum Computing

Quantum computing studies investigate the application of quantum optimization methods to transportation and logistics problems. QAOA and quantum annealing are among the approaches considered [7], [8], [10], [11].

E. Dynamic Transportation

Dynamic transportation studies consider situations in which transportation information changes over time. Such changes include traffic, demand, weather, and other operational conditions [16], [20].

XIX. Relationship to Smart Delivery Route Optimization

The selected literature provides several components relevant to smart delivery route optimization.

Big Data studies provide methods for handling transportation data and extracting information required for transportation optimization [1]–[3].

Logistics studies provide delivery-related optimization problems and application scenarios [4]–[6], [18], [20].

Vehicle routing studies provide the optimization problem required to determine delivery routes [2], [7].

Quantum computing studies provide emerging methods for solving combinatorial optimization problems [7], [8], [10], [11].

The reviewed literature therefore provides the basis for a hybrid approach in which transportation data are processed using Big Data methods and the resulting route optimization problem is investigated using quantum optimization.

The proposed project can therefore be represented as:

$$\text{Transportation Data} \rightarrow \text{Big Data Processing} \rightarrow \text{Route Optimization} \rightarrow \text{QU} \tag{2}$$

XX. Research Gaps

The reviewed literature identifies several areas that require further investigation.

A. Scalability

Vehicle routing problems become increasingly difficult as the number of customers, vehicles, and constraints increases. Quantum-based routing studies have also been evaluated on limited problem sizes because of current computational limitations [7].

B. Real-Time Transportation Information

Traffic, customer demand, weather, and vehicle conditions can change during delivery operations. Studies on dynamic routing and transportation demonstrate the importance of updated information for route planning [2], [16], [20].

C. Big Data and Quantum Integration

The selected literature contains studies addressing Big Data transportation analysis and studies addressing quantum optimization. Further work is required to integrate the two areas into a complete smart delivery route optimization framework.

D. Quantum Hardware Limitations

Quantum transportation studies are affected by the limitations of current quantum hardware and the size of problems that can be processed. These limitations influence the practical application of quantum optimization to large delivery networks [7], [9].

E. Dynamic Route Re-Optimization

Delivery routes may need to be recalculated when traffic conditions, customer requirements, or environmental conditions change. Efficient dynamic re-optimization remains an important research direction [2], [16], [20].

XXI. Discussion

The reviewed literature demonstrates the development of transportation optimization from data-driven logistics approaches toward quantum-based optimization.

Big Data studies provide methods for processing large volumes of transportation information. Logistics and routing studies provide the application problems and optimization requirements. Transportation studies demonstrate the importance of dynamic information such as traffic and demand.

Quantum computing provides an emerging approach for combinatorial optimization. QAOA is particularly relevant to the Vehicle Routing Problem because it can be formulated as an optimization problem using binary variables and QUBO representations.

The literature therefore provides the main research components for a hybrid Big Data and quantum computing approach to smart delivery route optimization.

However, the integration of transportation data processing and quantum optimization requires consideration of problem size, dynamic data updates, constraint handling, quantum hardware limitations, and solution validation.

XXII. Conclusion

This literature survey reviewed 20 selected research papers related to Big Data, logistics, route optimization, quantum computing, and dynamic transportation.

The reviewed studies demonstrate the use of Big Data analytics for transportation information processing and logistics optimization. Vehicle routing studies address the determination of delivery routes under different transportation constraints. Quantum computing studies investigate emerging approaches such as QAOA and quantum annealing for transportation optimization.

The literature provides a basis for combining Big Data transportation analysis with quantum optimization. In the proposed research direction, transportation information can be processed using Big Data techniques, followed by formulation of the delivery routing problem as a QUBO model and investigation using QAOA.

Future research can investigate larger delivery problems, dynamic route re-optimization, real-time transportation information, and hybrid classical-quantum approaches for practical smart delivery route optimization.

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